

Creating and Managing Users in Linux



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Overview



Welcoming Users to Linux

- Creating users with useradd
- Understanding /etc/passwd and getent
- Adjusting defaults
 - /etc/default/useradd
 - /etc/login.defs
- Populating home directories
- Modifying and deleting users
- Using find to locate files owned by deleted users
- Understanding the GECOS files and the command adduser
- Managing groups

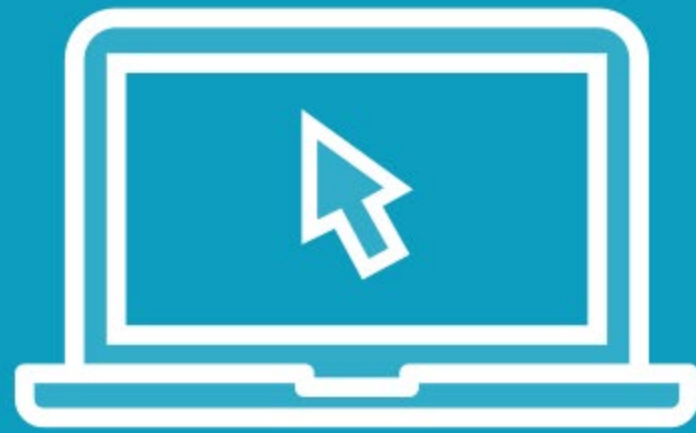


```
$ sudo useradd user1
$ sudo useradd --help
$ tail -n1 /etc/passwd
$ getent passwd user1
$ man 5 passwd
$ id user1
$ ls /home
$ sudo useradd -m -s /usr/bin/bash -G sudo user2
```

Creating Users

New users can be created using the command `useradd`. We need to elevate our permissions using `sudo`. An account will be created by using the defaults but we may prefer to add options as we will see

Demo



Creating Users

- Simply without options
- Adding options



Useradd Defaults

`/etc/default/useradd`

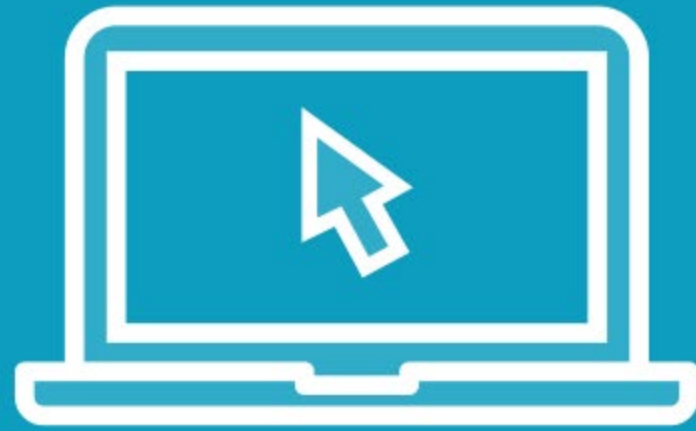
**We might change the default
shell to bash**

`/etc/login.defs`

**Here, we may choose to
automate home directory
creation**



Demo



Let's Investigate the Defaults:

- `/etc/default/useradd`
- `/etc/login.defs`
- create new users

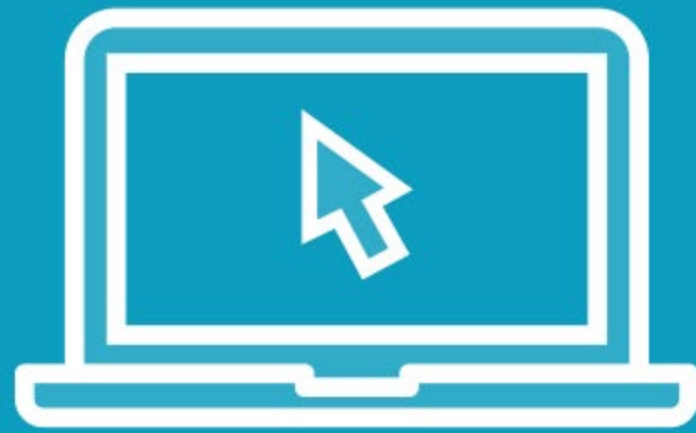


```
$ ls -la /etc/skel
$ sudo ln -s /usr/share/doc /etc/skel/doc
$ sudo useradd user4
$ ls ~user4
```

The Skeleton Directory

User home directories are templated from the `/etc/skel` directory. We could specify a different skeleton directory with the option `-k` to `useradd`. Adding or changing content within the directory will affect new home directories but not existing ones.

Demo



Skeleton Directory:

- Home directory template
- Adding content

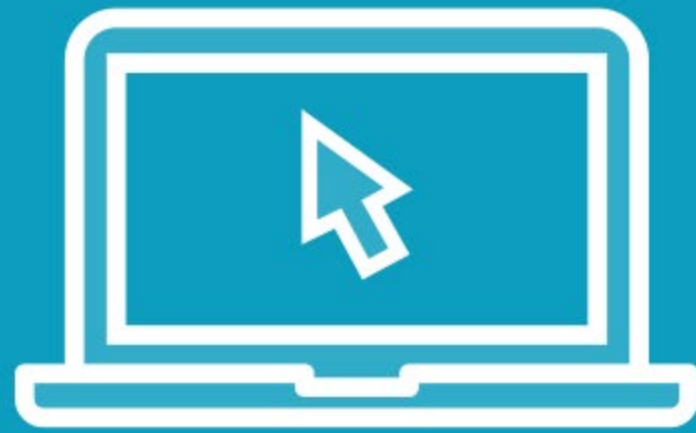



```
$ sudo usermod -G staff user2
$ id user2
$ sudo usermod -aG admin,sudo user2
$ id user2
$ sudo userdel -r user5
$ sudo userdel user3
$ ls /home
$ sudo find /home /var -nouser
$ !! -delete
```

Modify and Delete Users

We also have the `usermod` and `userdel` commands to further manage our users. Be aware that if we modify the group membership we will overwrite the current groups unless we append. When deleting accounts, we can also delete system created user files with `-r`.

Demo



Managing Users:

- modify users
- delete users
- search for files with no user owner



GECOS Field



General Electric Comprehensive Operating System (GECOS). Later, GECOS became General Comprehensive Operating System (GCOS). Mainly, we call it the comments field.

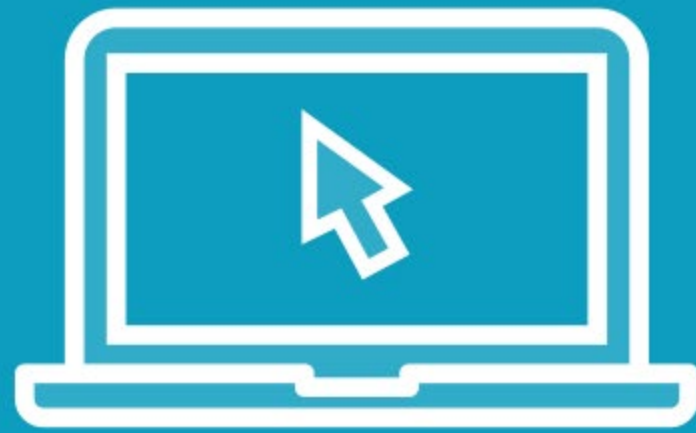


```
$ chfn  
$ sudo adduser user6  
$ sudo apt install -y finger  
$ finger user6
```

Finger Information

If the GECOS field is used correctly, we can store items like the office number, the work phone and home phone. Collectively we can refer to this additional data as "**finger data**" we can set with the **chfn** command. If installed, we can read the data in a pretty format using the **finger** command. When creating users the **adduser** script will set finger data.

Demo



Working with the Comments or GECOS Field:

- chfn
- adduser
- finger

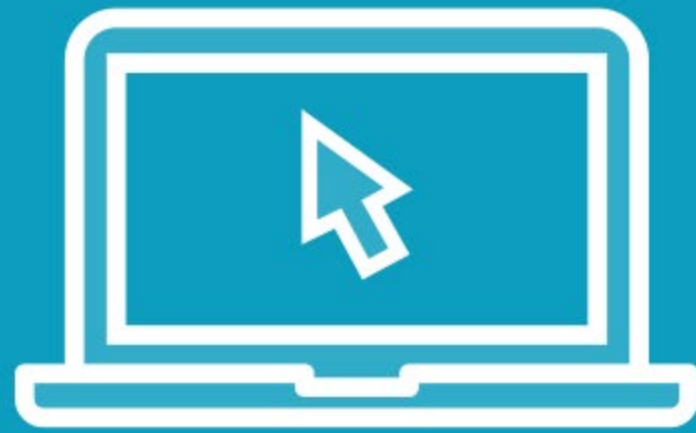



```
$ cat /etc/group
$ getent group sudo
$ sudo groupadd sales
$ sudo groupmod -n grp-sales sales
$ sudo groupdel grp-sales
```

Groups

Groups are created in the `/etc/group` file. We can view the file or use the `getent` command to list groups. We can rename a group using `groupmod` and the option `-n`. `Groupdel` is used to delete the group

Demo



Managing User Groups:

- groupadd
- groupmod
- groupdel



Summary



Managing Users and Groups

- /etc/passwd
- /etc/group
- getent [passwd | group]
- useradd/mod/del
- groupadd/mod/del
- find /home -nouser
- chfn
- adduser
- finger



Managing User Passwords and Authentication in Linux

